

Q1 a) The total transmission delay from source to destination: $T_1 + d_1 + T_2 + d_2$
 where $T_1 = \left(\frac{L}{R_1}\right)$, which means that it is the delay on 1st link to get all L length bits, similarly T_2 is the delay on 2nd link to get all L length bits on it (transmitting on them).
 The d 's are the delays while propagating from 1 to another. d_1 being on first link, d_2 for travelling the distance of link 2.
 So the original equation would be decomposed into: $\frac{L}{R_1} + d_1 + \frac{L}{R_2} + d_2$.

b) Since $R_2 \ll R_1$, that means the delay by second link gets more since $T_2 \propto \frac{1}{R_2}$, so T_2 will increase, and R_1 increasing means T_1 will be negligible compared to T_2 , hence now the total delay depends on: $\frac{L}{R_2} + d_1 + d_2$

Q2 a) HFC (Hybrid ^{Cables of} Fiber Coaxial) has TV based infrastructure and shares the bandwidth available by it, max speed given by it is a 'possible' speed, not guaranteed since it is shared. However DSL (Digital subscriber line) has telephone lines (which are basically twisted copper lines) give the bandwidth to each person separately and private/dedicated for them. Max speed ^(100 mpps) here now depends on the transmission lines instead of users. Reference of RJ Page 13

b) Since the people in HFC network are all sharing the network, through the same coaxial cable, the traffic is huge, due to which delays occur ^{on 100 mpps speed} by queues forming. However since DSL have a specific line for each customer it doesn't get effected as long as the length is greater from the provider's office.

Q3 a) Headers at application: Here it is as HTTP request header and tells ~~which server~~ ^{being requested} the specific website, carrying information to let it know the browser type as well as language preference in its format.

Headers at transport: Here it is as TCP header, having source and destination ports to make sure that the request is reaching the correct application on server.

Headers at Network: Here it is as IP header, giving IP address which are logical for a specific HTTP request, so that it can reach the destination server even by multiple routes.

Headers at Link: Here it is as Ethernet header, giving the MAC address and can go between connected or linked nodes. Reference 24 and 25 above and below respectively.

b) As the router's function is to deliver the packets to the host in its destination (within the range or not), it is not dependent on server or application. So Transport and application layer will be ignored, no need for ports ~~as well~~ as well.

It ~~examines~~ ^{network} examines link (as to know which host to go by MAC address) and ~~for~~ (for destination's IP address).

c) By having segmentations, even if one packet fails to reach destination due to being corrupted, then entire image would not need a retransmission and other packets will go on (if UDP).

Disadvantage will be that having a lot of packets can form overhead and increase the delay due to a lot of traffic.

Q4 a) no. of users = $\frac{\text{total link speed}}{\text{speed usage per customer}} = \frac{10 \times 10^9}{100 \times 10^6} = \frac{1000}{100} = \boxed{10}$ users can be supported simultaneously.

- b) Since in packet switching, no fixed circuit is there. Hence when the a user is not active, the system has more space and free to share packets on no chance of failure, usage being dynamic as well.

Q5 a) The analogy can be explained by mapping packet as a car, since car carries passengers, packets carry data, travelling independently through network. The driver of the car would be packet header which knows where to take the car with passengers (^{contains} IP address ^{for the direction}). Upon every intersection, driver would see where to go next, so intersection being a router. And lastly, the final destination ^(direction) would be the endpoint for packets which will be a host that required them.

- b) An entity setting up a fake sign on intersection corresponds to network layer since it relates to routers/routing. This way packets ~~can~~ be sent distracted from the real direction, can be referred as hijacking, also ^{this} may change content ~~in~~ in the packet by someone present in the wrong path receiving them. Can degrade performance due to longer routes or denial of service if packets fail to reach, as well.

- c) Among the three principles, Authentication is primarily being violated, since the entity ensures that the path ^{is the correct one} ~~is~~ ^{provided by it} ~~where~~ they want to go, hence giving false information, confidentiality is keeping information private ~~which~~ which is not the case here, some form of integrity can also be violated as if the packets are altered but this is not the primary case.

Reference R20 and R28.

Q6 P5:- a) there will be two delays, one while propagating from ^{starting} booth till ending booth, other while the cars are at 'each' booth.

$$\text{propagation delay} = \frac{\text{distance in propagation}}{\text{speed with which caravan travelled}} = \frac{175 \text{ km}}{100 \text{ km/h}} = 1.75 \text{ hours}$$

■ toll booth delay :- 10 cars in caravan

1 car = 12 seconds service delay.

120 seconds delay for caravan.

since there are 3 toll booths, $120 \times 3 = 360 \text{ seconds}$

1.75 hours $\rightarrow (1.75 \times 3600) \text{ seconds} \rightarrow 6300 \text{ seconds}$

$$\text{Total delay} = 6300 + 360 = 6660 \text{ seconds} \Rightarrow \frac{6660}{3600} = 1.85 \text{ hours}$$

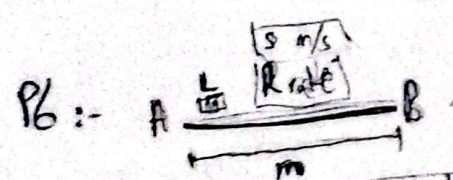
b) having 8 cars = less time in service (at toll booth).

propagation delay remains the same (6300 seconds)

1 Toll booth delay = $8 \times 12 = 96 \text{ seconds}$

3 toll booth delay = $96 \times 3 = 288 \text{ seconds}$

$$\text{Total delay} = 6300 + 288 = 6588 \text{ seconds} \Rightarrow \frac{6588}{3600} = 1.83 \text{ hours}$$



a) $d_{\text{prop}} = \frac{m \rightarrow \text{distance}}{S \rightarrow \text{speed}}$

b) $d_{\text{trans}} = \frac{L \rightarrow \text{bits}}{R \rightarrow \text{bits/s}} = \frac{L}{R} \text{ seconds of delay}$

c) no queue / service delays = $\text{Total} = \frac{L}{R} + \frac{m}{S}$

d) since transmission time is the time it takes to put the packet/bit onto link and $\frac{L}{R}$ means having L bits on link with rate R (between A and B) here last bit of L just left A and ^{now the} onto link to get to B.

e) As propagation delay is greater than transmission delay, ^{so} as all bits are transmitted to link from A to B, the first bit that left way earlier hasn't still reached B and is still on the link

D) Now as propagation delay is smaller than d_{trans} , hence till the last bit would have been transmitted (till d_{trans}) the first one would have been already at B, waiting to transmit, or something.

g) $d_{prop} = d_{trans}$

$$\frac{m}{s} = \frac{L}{R}$$

$$m = \frac{1000 \times 8}{10 \times 10^6} \times 2.5 \times 10^8 = 1000 \times 8 \times 25 = \boxed{200000 \text{ bytes}}$$

18:- a) Final hops: 94, 96, 94

hr1: mean = 94.67ms, s.d = $\sqrt{1.335} = 1.16\text{ms}$

hr2: 99, 97, 98
mean = 98ms, s.d = 1ms

hr3: 93, 93, 93
mean = 93ms, s.d = 0ms

b) In all 3 times, there were a total of 15 hops.
Which was 14 routers and 1 for destination server.

c) From 1-5 hop, no ISP change has; it changes in 6 hop, and then in 7th as well. So ~~total~~ total Three public networks, access ISP, transit exchange, and then destination ISP.
largest delay at transit interface

d) Now for new continent's transit.

hr1: 83, 76, 73
mean = 78.33, s.d = 4.04ms $\rightarrow$ 10 routers, 1 dest

hr2: 84, 84, 84, s.d = 0ms $\rightarrow$ 16 routers, 1 dest

hr3: 97, 99, 97, s.d = 1.16ms $\rightarrow$ 16 routers, 1 dest

four public networks.

Access ISP 1st hop.

Core (Transit) hop 10

Telia (Transit) hop 12

destination network hop 13-17

f) Now as propagation delay is smaller than transmission delay, and ~~data~~ ~~all~~ ~~bits~~ ~~are~~ ~~being~~ ~~transmitted~~, first bit was 0 since rest of the ~~bits~~ were

P24 :- $50 \text{ TB} = 50 \times 10^{12} \times 8 = 4 \times 10^{14} \text{ bits}$

$$d_{\text{trans}} = \frac{\text{size in bits}}{\text{rate}} = \frac{4 \times 10^{14}}{100 \times 10^6} = 4 \times 10^6 \text{ seconds}$$

$$\frac{4 \times 10^6}{86400} = 46.3 \text{ days to transmit 50TB over 100 Mbps link}$$

P31 :- a) $d_{\text{trans}} = \frac{L}{R} = \frac{10^6}{5 \times 10^6} = 0.2 \text{ seconds to first switch}$

3 links, $T_{\text{segment}} = 3 \times \frac{L}{R} = 0.6 \text{ seconds, total to destination}$

b) $t = 2 \text{ ms}$ ~~can start sending packet 1~~ ~~onto link 2~~ while source sends packet 2, packet 2 then fully received at switch 1 at $t = 0.004 = 400 \text{ ms}$ (200+200)

c) first packet to reach the destination (3 links) = $3 \times 200 \text{ ms} = 600 \text{ ms}$
since first arrived, now 99 remains.

$$\text{so Total time} = (600) + (99 \times 200) = \boxed{204 \text{ ms}}$$

No segmentation in a) = 600 ms, with segmentation = 204, speedup from as multiple links used in parallel.

d) reasons to use message segmentation could be to enable multiplexing, error recovery, reducing buffering on 1 link.

e) Drawbacks of message segmentation will increase overhead from headers due to more packets, higher processing rate in route with reassembly complexity increasing at host end.

Bonus:-

a) Total end to end delay for a single packet is sum of all ~~transmission~~ ^{delays}.

$$d_{\text{Trans}} = \frac{(1500 \times 8)}{\text{All different bandwidths.}}$$

$$= (1500 \times 8) \times \left(\frac{1}{10^9} + \frac{1}{10^9} + \frac{1}{100 \times 10^9} + \frac{1}{50 \times 10^9} + \frac{1}{100 \times 10^9} + \frac{1}{100 \times 10^9} + \frac{1}{10 \times 10^9} \right) \rightarrow \text{rate}$$

$$= 0.0258 \text{ ms}$$

$$d_{\text{prop}} = \left(\frac{10}{2 \times 10^5} + \frac{5}{2 \times 10^5} + \frac{800}{2 \times 10^5} + \frac{6500}{2 \times 10^5} + \frac{200}{2 \times 10^5} + \frac{150}{2 \times 10^5} + \frac{20}{2 \times 10^5} \right) = 38.42 \mu\text{s}$$

$$\text{process/service delay} = 0.1 + 0.5 + 0.2 + 0.2 + 2 + 0.3 + 0.1 + 0.2 = 3.4 \text{ ms}$$

$$\text{Total} = 0.0258 + 38.42 \mu\text{s} + 3.4 = \boxed{41.9008 \text{ ms}}$$

b) bottleneck (lowest) $\Rightarrow 1 \text{ Gbps}$

$$\text{Total transmission time} = \frac{80 \times 10^9 \text{ bits}}{1 \times 10^9 \text{ bps}} = 80 \text{ seconds}$$

$$\text{Total delay} = 41.9008 + 80 = \boxed{121.9008 \text{ seconds}}$$

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